

# Rahul Sheshanarayana

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in [linkedin.com/in/rahul-sheshanarayana/](https://linkedin.com/in/rahul-sheshanarayana/)    🎓 [scholar.google.com/citations?user=Zhd0xIMAAAAJ](https://scholar.google.com/citations?user=Zhd0xIMAAAAJ)

PhD candidate focused on designing safer molecules and remediating PFAS contamination. Developed generative models, scalable ML systems, and molecular simulation tools with demonstrated impact across sustainability and materials science. 7+ publications in leading journals including *Advanced Science*, *Nature Communications*, and *Digital Discovery*, with multiple journal covers and an Editor's Pick.

## Education

### PhD, Cornell University

Aug 2024 - Present

Major - Systems Engineering

Thesis title: Thermochemistry-aware Machine Learning for Sustainable Molecular Design: Representation Learning, Radical Design, and PFAS Remediation

Thesis advisor: Dr. Fengqi You

### MS, Cornell University

Aug 2022 - May 2024

Major - Chemical Engineering

Thesis title: Probing Ion Effects In Nanoconfined Aqueous Electrolytes: A Molecular Dynamics Study Using Neural Network Potentials

Thesis advisor: Dr. Shuwen Yue

### BTech, Indian Institute of Technology (IIT) Roorkee

Aug 2018 - May 2022

Major - Chemical Engineering (specialization in Polymer Science), Minor - Applied Mathematics

Thesis title: A Kinetic Model for the Direct Thermal Liquefaction of Pine Wood

Thesis advisor: Dr. Shushil Kumar

## Research Experience

### Department of Systems Engineering, Cornell University

Aug 2024 - Present

- Designed a bond dissociation energy (BDE)-conditioned transformer for generative molecular design, achieving distributional control over radical bond strengths with 81-94% validity and 84-92% novelty across BDE targets.
- Achieved up to 90%  $R^2$  improvement in molecular property prediction by distilling SchNet and DimeNet++ into  $2\times$  smaller student GNNs, across QM9, ESOL, and FreeSolv datasets.

### Department of Chemical Engineering, Cornell University

Aug 2022 - July 2024

- Trained DFT-accurate neural network potentials via active learning on HPC clusters for nanoconfined electrolytes, reaching force RMSEs  $\leq 0.07$  eV/Å and enabling fast MD simulations.
- Identified ion-specific interfacial behavior in confined electrolytes;  $K^+$  showed  $3\times$  stronger adsorption and  $2\times$  faster diffusion than  $Na^+$ , trends absent in classical force fields.

### Department of Chemical Engineering, IIT Roorkee

Aug 2021 - Aug 2022

- Developed a kinetic model for pine wood liquefaction, finding distillate formation nearly  $2\times$  faster than heavy residue, with secondary reactions negligible.
- Demonstrated that adding 20 wt% water more than doubled primary reaction rates, with guaiacol:water = 8:1 giving the highest overall conversion.

### Department of CSE, Jadavpur University

Jan 2021 - Aug 2022

- Boosted vehicular smoke detection accuracy by up to 12% mAP across 3 public datasets using a  $\lambda$ -attention transformer head on a YOLOv5 backbone.
- Increased training coverage by  $5\times$  with a dual-level synthetic smoke generation pipeline (mask patterns + filtering) to overcome limited real-world data.

### Department of Chemical Engineering, Indian Institute of Science

Nov 2020 - Apr 2022

- Achieved 97%  $R^2$  for nanopore formation probability and 95%  $R^2$  for formation time prediction by training a two-stage ML framework on 20,840 unique graphene nanopore structures.
- Quantified the effect of 18 structural features (e.g., rim atoms, shape factor, symmetry) on pore formation kinetics using SHAP-based feature importance, providing interpretable physical insights.

## Publications

Sheshanarayana, R., You, F. (2026). Harnessing homolytic bond energetics to steer inverse radical design.

Verma, A.M., Chaturvedi, S., Paul, S., Nandi, S, **Sheshanarayana, R.**, Santhosh, K., Valavarasu, G., Dukkupati, A., Gwie, C.G., Moo, P.Y., Ng, C.Q.J., Amrute, A., Govind Rajan, A. (2026). Data-driven massive reaction networks reveal new pathways underlying catalytic CO<sub>2</sub> hydrogenation. *Nature Communications (Accepted in Principle)*.

**Sheshanarayana, R.**, You, F. (2025). Rethinking Retrosynthesis: Curriculum Learning Reshapes Transformer-Based Small-Molecule Reaction Prediction. *J. Chem. Inf. Model.*, 65(20), 11047-11063. **Supplementary Journal Cover.**

**Sheshanarayana, R.**, You, F. (2025). Molecular Representation Learning: Cross-domain Foundations and Future Frontiers. *Digital Discovery*, 2298. **Journal Inside Front Cover.**

**Sheshanarayana, R.**, You, F. (2025). Knowledge Distillation for Molecular Property Prediction: A Scalability Analysis. *Advanced Science*, 2503271. **Journal Frontispiece.**

**Sheshanarayana, R.**, Kumar, S. (2025). A kinetic model for the direct thermal liquefaction of pine wood. *Biomass Conversion and Biorefinery*, 15(17), 23795-23803.

Kundu, S., Maulik, U., **Sheshanarayana, R.**, Ghosh, S. (2022). Vehicle Smoke Synthesis and Attention-based Deep Approach for Vehicle Smoke Detection. *IEEE TIA*, 59(2), 2581-2589.

**Sheshanarayana, R.**, Govind Rajan, A. (2022). Tailoring nanoporous graphene via machine learning: Predicting probabilities and formation times of arbitrary nanopore shapes. *J. Chem. Phys.*, 156(20). **Editor's pick.**

## Skills

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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scientific domains  | Thermochemistry, reaction modeling, polymer science, nanoporous materials, confined electrolytes, statistical mechanics, computational fluid dynamics, linear/non-linear programming, goal programming, control theory. |
| Computational tools | Density functional theory, Monte Carlo simulations, molecular dynamics simulations, statistical optimization, machine learning, deep learning.                                                                          |
| Software            | Python, R, DPGEN, DeepMD, Quantum Espresso, ORCA, LAMMPS, MATLAB, L <sup>A</sup> T <sub>E</sub> X, VESTA, VMD.                                                                                                          |
| ML packages         | PyTorch, PyTorch Geometric, Hugging Face Transformers, Scikit-learn, ASE, RDKit, Optuna, Weights & Biases, TensorFlow, Pandas, NumPy, Matplotlib.                                                                       |

## Teaching

**Graduate Teaching Assistant (Investigative Biology Lab)** **Aug 2023 - May 2025**

Taught 35 undergraduates per semester in Investigative Biology (BIOG 1500) at Cornell, covering experimental design, hypothesis testing, and statistical analysis through hands-on labs.

**Graduate Teaching Assistant (Fundamentals of Physics II Lab)** **Feb 2023 - May 2023**

Guided students through lab experiments ranging in topics from electricity and magnetism to optics.

## Academic Awards

**Best Bachelor's Thesis Award** **Aug 2022**

Received for developing a kinetic model for the direct thermal liquefaction of pine wood, later published in *Biomass Conversion and Biorefinery*.

**Editor's Pick - Journal of Chemical Physics** **Jan 2022**

Awarded for "Tailoring nanoporous graphene via machine learning: Predicting probabilities and formation times of arbitrary nanopore shapes"- selected as Editor's Pick in the *Journal of Chemical Physics*.